

RESEARCH ARTICLE

Prompt Engineering Versus Model Selection for Cognitive Accessibility in Large Language Models: An Empirical Study

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ABSTRACT The promise of large language models (LLMs) for individuals with learning disabilities and cognitive impairments is significant, yet their practical accessibility remains limited by readability barriers. In improving output accessibility, both model selection and prompt engineering offer potential levers, but their relative importance remains unquantified. This study conducted a systematic investigation of prompt engineering versus model selection for enhancing cognitive accessibility of responses in four leading LLMs (Llama-3.3-70B, Llama-3.1-8B, GPT-4o, Claude-Sonnet-4.5). Three prompt engineering approaches were tested: Baseline (task only), Structured (organizational guidance), and Cognitive-Explicit (accessibility constraints grounded in working memory theory), across two task types (procedural and conceptual), resulting in a $3 \times 2 \times 4$ factorial experiment ($N = 240$ responses), with readability quantified as Flesch Reading Ease scores. Prompt engineering accounted for 71.9% of variance ($\eta^2 = 0.719$, $p < 0.001$) versus only 15.8% from model selection ($\eta^2 = 0.158$, $p < 0.001$). Cognitive-Explicit prompts improved readability by 37.64 points (Cohen's $d = 2.49$), reducing sentence length by 47% with minimal trade-off in lexical diversity. Task type also influenced accessibility, with procedural tasks yielding significantly higher readability than conceptual explanations ($p < 0.001$, $\eta^2 = 0.231$). Models exhibited differential responsiveness, reflected in a marginally significant Model $\times$ Prompt interaction ($p = 0.068$, $\eta^2 = 0.052$). Claude-Sonnet-4.5 achieved the largest gain (+43.41 points, +128%) and Llama-3.1-8B the smallest (+31.36 points, +64%). These findings establish prompt engineering as the dominant factor for cognitive accessibility in LLMs and offer actionable guidance for practitioners prioritizing accessibility in AI deployments for cognitive disability populations.

INDEX TERMS Accessibility design, AI ethics, cognitive accessibility, large language models, learning disabilities, prompt engineering, readability.

I. INTRODUCTION

Large language models (LLMs) are rapidly finding applications in domains as diverse as customer service, education, healthcare, and assistive technologies [1], [2]. Especially, they offer a promising avenue for bridging information access gaps and thereby supporting individuals with learning disabilities and cognitive impairments such as dyslexia, intellectual disabilities, and autism spectrum disorder [3], [4]. However, current LLMs are also fundamentally mismatched with this

purpose, being optimized for fluency, coherency, and factual accuracy rather than cognitive accessibility [5], [6]. That is, the output of LLMs frequently features lengthy sentences, complex syntax, and dense vocabulary that are not readily encompassed by the working memory and processing capacities of many users with cognitive disabilities [7], [8].

In addition, accessibility is further challenged by the rapid proliferation of LLM options. Amidst this profusion, organizations deploying AI systems must make two core decisions: (1) choice of model (e.g., GPT-4o, Claude-Sonnet, Llama-3.3-70B) and (2) their approach to prompt engineering so as to elicit the desired behavior [9], [10]. The respective

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answers to these decisions are well-recognized as key influences on output quality; however, how they relate to cognitive accessibility remains undetermined. The practical implications of this knowledge gap are significant, as the element with greater influence on accessibility outcomes should be prioritized. That is, if model selection is more determinant of accessibility, model procurement strategies should be prioritized; but if prompt engineering predominates instead, then prompt design, prompt libraries, and user education should receive focus [11], [12].

To date, accessibility improvements have been explored from each direction in isolation, considering either model selection [13], [14] or prompt engineering [15], [16]. No direct comparisons of relative importance yet exist. In addition, physical disabilities (e.g., vision, hearing) have been the focus of attention rather than cognitive accessibility [17], [18]. What work there is on cognitive accessibility in LLMs has relied on anecdotal examples or small case studies [19]. The current lack of controlled experimental designs leaves critical questions answered, namely: *How much* does prompt engineering improve readability compared to model selection? Which factor should practitioners prioritize? Do these effects interact, such that certain models are more responsive to accessibility-focused prompts?

The present work conducts a systematic factorial experiment to address these questions, comparing prompt engineering and model selection as mechanisms for enhancing the cognitive accessibility of LLMs. Three prompt engineering approaches are considered: Baseline (task-only queries), Structured (generic organizational guidance), and Cognitive-Explicit (specific accessibility constraints grounded in working memory theory and cognitive load principles [20], [21]). These different approaches are evaluated for two task types (procedural vs. conceptual) in four leading LLMs (Llama-3.3-70B, Llama-3.1-8B, GPT-4o, Claude-Sonnet-4.5). Flesch Reading Ease is taken as the primary accessibility metric [22], and the relative contributions of prompt engineering (ηp^2), model selection ($\eta \mu^2$), and their interaction are quantified while controlling for task type effects.

The results of this evaluation reveal that prompt engineering has a 4.6-fold larger effect, accounting for 71.9% of variance in readability ($\eta p^2 = 0.719$), versus just 15.8% for model selection ($\eta \mu^2 = 0.158$). With regard to different engineering approaches, prompts following the Cognitive-Explicit design improved Flesch Reading Ease by 37.64 points (Cohen's $d = 2.49$) and reduced sentence length by 47%. Significant accessibility gains were seen with this approach in all four tested models. The marginally significant Model $\times$ Prompt interaction ($p = 0.068$) indicated differential responsiveness of models to accessibility prompts. The most responsive model was Claude-Sonnet-4.5 (+43.41 points, +128% improvement), and the least Llama-3.1-8B (+31.36 points, +64% improvement). These results provide the first quantitative evidence that prompt engineering is the dominant lever for cognitive accessibility in LLMs, offering actionable guidance for practitioners.

The remainder of this paper is organized as follows: Section II reviews related work on LLM accessibility, prompt engineering, and cognitive disability. Section III details our experimental design, prompt engineering approaches, and measurement framework. Section IV presents statistical analyses of main effects, interactions, and effect sizes. Section V discusses implications for accessibility design, limitations, and future research directions. Section VI concludes with practical recommendations for organizations deploying LLMs for cognitive disability populations.

II. RELATED WORK

A. COGNITIVE ACCESSIBILITY AND LLMs

Cognitive accessibility refers to design practices that reduce cognitive load, support working memory limitations, and accommodate diverse information processing needs [23], [24]. Overall, an estimated 15-20% of people worldwide live with some form of learning or cognitive disability [25], [26], including dyslexia, intellectual disabilities, ADHD, and autism spectrum disorder. This population is disproportionately affected by barriers to cognitive accessibility, such as dense paragraph structures, complex syntax, lengthy sentences, and unfamiliar vocabulary [27], [28].

LLMs offer intriguing possibilities as assistive tools for people with cognitive disability, which recent research has begun to explore. GPT-3 has been demonstrated capable of simplifying medical information for individuals with intellectual disabilities, though improvements to readability were inconsistent [29]. Prompt-based text simplification can improve output comprehension by dyslexic readers [30]; however, a systematic comparison of prompt approaches and model architectures is yet lacking. A conversational AI system has been developed specifically for individuals with autism and received positive feedback from users; however, no quantitative measures of accessibility were collected [31]. Thus, while these reports support the potential of LLMs in improving cognitive accessibility, controlled experimental studies that comprehensively compare model and prompt effects remain needed.

B. PROMPT ENGINEERING FOR ACCESSIBILITY

Model fine-tuning is one means of shaping LLM behavior; prompt engineering has emerged as another [32], [33]. Engineering strategies aimed at improving accessibility include incorporating explicit readability instructions [34], few-shot examples of simplified text [35], and role-based prompts (e.g., "Explain like I'm five") [36]; however, these strategies lack theoretical grounding in cognitive science, and implementations have largely been undertaken in an ad hoc manner, without systematic evaluation.

More recently, efforts have been made towards formalizing such accessibility-oriented prompt engineering. "Cognitive scaffolding prompts," proposed by Zhang et al. [37], have been shown to improve comprehension in preliminary tests; this strategy involves chunking information, providing concrete examples, and minimizing jargon. Zero-shot, few-shot,

and chain-of-thought prompting were compared by Williams and Park [38], with example-based approaches being outperformed by explicit readability constraints. No study has yet quantified the relative importance of prompt engineering versus model selection for output accessibility, nor has a systematic comparison of prompt design philosophies been undertaken (e.g., generic organizational guidance vs. explicit cognitive constraints).

C. MODEL SELECTION AND OUTPUT CHARACTERISTICS

The various LLM architectures differ in the character of their outputs, which may have an influence on cognitive accessibility. Outputs of larger models (e.g., GPT-4, Claude-3.5) tend to be more context-rich and verbose, whereas those generated by smaller models (e.g., Llama-3.1-8B) tend to be more concise [39], [40]. Models may also be tuned for particular objectives in training, with consequent impacts on output characteristics; for example, base models prioritize fluency, but instruction-tuned models put import on helpfulness and harmlessness [41], [42].

Only limited empirical comparisons of model readability are presently available. In an analysis of five LLMs, Patel et al. [43] observed significant readability differences in base output, but considered those differences to result from training data rather than model architectures. In a comparison of base versus instruction-tuned models, Rodriguez and Kim [44] found the latter to produce more accessible outputs, but did not control for prompt engineering. Together, these findings hint at model selection having an effect on output accessibility, but its importance in comparison to prompt design remains to be determined.

D. RESEARCH GAP

To date, the influence of model selection and prompt engineering on output accessibility have each been investigated in isolation. In addition, studies have largely used informal prompts, been conducted on small samples, and lacked control groups. There is no extant systematic comparison of prompt engineering versus model selection in a study using factorial design; no quantification of effect size (η^2) for model, prompt, and interaction effects; no study using prompt design strategies grounded in cognitive science theory (working memory, Cognitive Load Theory); and no evaluation of accessibility across different task types (procedural vs. conceptual) in a range of LLMs (Llama, GPT, Claude).

The present study employs a rigorous experimental design in addressing these questions and filling the gaps they represent. It thereby provides the first quantitative evidence on the relative importance of prompt engineering versus model selection for cognitive accessibility in LLMs.

III. METHODOLOGY

A. EXPERIMENTAL DESIGN

Three independent variables were considered: prompt type, task type, and model. Prompt type comprised three levels:

Baseline, Structured, and Cognitive-Explicit; task type two levels: Procedural and Conceptual; and model four levels: Llama-3.3-70B, Llama-3.1-8B, GPT-4o, and Claude-Sonnet-4.5. Together, these made for a $3 \times 2 \times 4$ factorial study design with a total 24 experimental conditions. Each condition was replicated 10 times with unique task queries, for a total $N = 240$ responses. This design provides 95% power to detect an effect of medium size ($f = 0.25$) at $\alpha = 0.05$ for main effects and interactions [45].

One dependent variable was utilized, Flesch Reading Ease (FRE) [22], which quantitatively represents the cognitive accessibility of a text. This measure is computed as:

$$\text{FRE} = 206.835 - 1.015 \times (\text{total words} / \text{total sentences}) - 84.6 \times (\text{total syllables} / \text{total words})$$

Values range from 0-100, with higher scores indicating greater readability. On this scale, an 8th-9th grade reading level (standard) is represented by a score of 60-70, and a 6th grade level (easy) by a score of 80-90.

To assess the robustness of the results, we supplemented FRE with 14 additional readability metrics (e.g., Flesch-Kincaid Grade Level, SMOG Index, average sentence length) [46].

B. PROMPT ENGINEERING APPROACHES

Baseline (control) prompts consisted of the plain task query with no readability guidance. For example, “What is cloud storage?” The two experimental approaches we designed, Structured and Cognitive-Explicit, were respectively grounded in working memory theory [20] and Cognitive Load Theory [21]. Structured prompts incorporated generic organizational guidance but no specific cognitive constraints. As a case in point, “Explain cloud storage using simple words and short sentences.”

Distinct from the first two types, Cognitive-Explicit prompts contained explicit accessibility constraints targeting working memory capacity (7 ± 2 chunks [20]), lexical familiarity, sentence length (< 15 words), and chunking. For example, “Explain cloud storage for someone with learning difficulties. Use: (1) simple everyday words, (2) sentences under 15 words, (3) one main idea per sentence, (4) concrete examples, (5) bullet points or numbered lists.” This category of prompts operationalizes cognitive accessibility principles. Namely, a sentence length < 15 words reduces working memory load [47]; use of simple, everyday words minimizes lexical processing demands [48]; encapsulating only one idea within a sentence prevents cognitive overload from embedded clauses [49]; use of concrete examples supports schema activation for abstract concepts [50]; and providing output in lists chunks information for easier encoding [51].

Complete templates for all prompt types are provided in Appendix A.

C. MODEL SELECTION

Four leading LLMs were selected for use in this study, representing diverse architectures and parameter scales.

Llama-3.3-70B, from Meta, is instruction-tuned with 70 billion parameters, available under the Apache 2.0 license. Its smaller-scale alternative, Llama 3.1-8B, differs in using only 8 billion parameters. GPT-4o, from OpenAI, is a multimodal LLM optimized for speed and cost. Finally, Claude-Sonnet-4.5, from Anthropic, is a safety-focused constitutional AI.

Official APIs were employed in all cases: the Groq Python SDK v0.13 for Llama models, OpenAI Python SDK v1.x, and Anthropic Python SDK v0.39. The following standardized generation parameters were supplied: temperature = 0.7 (moderate creativity), max_tokens = 512 (sufficient for explanations), top_p = 1.0 (nucleus sampling disabled).

D. TASK DESIGN

Task queries were conceptualized around 4 domains relevant to everyday life for cognitive disability populations. These comprised: healthcare, for example medication instructions, symptom explanations, medical procedures; concerns of daily living, such as banking, transportation, household management; education, including academic concepts, study strategies, homework assistance; and employment and financial matters, such as job applications, workplace safety, budgeting. Fifteen tasks were created in each domain, for 60 unique queries total. Within each domain, about half (7-8) of the tasks were procedural in nature (“How do I...?”), intended to elicit step-by-step instructions. The other half were conceptual, requiring definitional explanations in answer. We hypothesized that the sequential, concrete structure of procedural tasks would produce outputs with higher baseline readability [52].

The complete task inventory is provided in Appendix B.

E. DATA COLLECTION

Data was collected in November 2024. For each of the 60 base queries, 3 responses were generated (one per prompt type) for each of the 4 models, yielding 240 total responses. All API calls included error handling, retry logic (max 3 attempts), and response validation (minimum 50 characters, maximum 512 tokens). Outputs were also subjected to the following quality control checks: response not empty or truncated; not an API error; and not a duplicate of a previous response. All 240 responses met these quality standards, yielding 100% usable data. The total API cost was \$2.10. To ensure reproducibility, responses were stored with associated metadata (model, prompt type, task type, timestamp). The complete response dataset is provided in Appendix C.

F. READABILITY MEASUREMENT

Flesch Reading Ease (FRE) served as the primary dependent variable on account of it being widely used in accessibility research, having validated grade-level equivalents, and aligning with cognitive load principles [22], [53]. An additional 14 metrics were computed as supplemental measures for robustness evaluation. These grouped into four types as follows:

TABLE 1. Descriptive statistics by model and prompt type.

Model	Baseline	Structured	Cognitive-Explicit	Marginal Mean
Llama-3.3-70B	49.97 (10.79)	84.45 (5.47)	86.09 (8.37)	73.51
Llama-3.1-8B	48.94 (8.88)	72.23 (6.01)	80.30 (6.48)	67.16
GPT-4o	42.61 (11.04)	75.55 (5.53)	82.27 (4.31)	66.81
Claude-Sonnet-4.5	33.93 (18.26)	71.27 (10.19)	77.34 (7.89)	60.85
Marginal Mean	43.86	75.88	81.50	67.08

- Grade Level: Flesch-Kincaid Grade, SMOG Index, Coleman-Liau Index, Gunning Fog
- Sentence Complexity: Average sentence length, average syllables per word
- Lexical Complexity: Difficult words (≥ 3 syllables), lexical diversity (type-token ratio)
- Structural Features: Bullet points, numbered lists, paragraph count

All metrics were computed using the Python textstat library v0.7.3 [54] with NLTK v3.8 for sentence tokenization [55]. The complete readability analysis results are provided in Appendix D.

G. STATISTICAL ANALYSIS

Main effects and interactions were assessed for statistical significance using three-way ANOVA. Independent variables consisted of the model (4 levels), prompt type (3 levels), and task type (2 levels), with FRE as the dependent variable. Effect size was quantified as partial eta-squared (η^2), interpreted per Cohen’s guidelines: small ($\eta^2 = 0.01$), medium ($\eta^2 = 0.06$), large ($\eta^2 = 0.14$) [56]. Tukey’s HSD was applied in post-hoc pairwise comparisons to control Type I error. Cohen’s d was computed for key contrasts (Baseline vs. Cognitive-Explicit). All statistical analyses were conducted in Python using scipy.stats v1.11 and statsmodels v0.14 [57], [58]. The complete ANOVA results and descriptive statistics are provided in Appendix E.

IV. RESULTS

A. DESCRIPTIVE STATISTICS

Upon evaluating all 240 responses, a clear effect of prompt type was observed, with Cognitive-Explicit prompts yielding substantially higher FRE than Baseline ($M = 81.50$ versus 43.86 , $+37.64$ points). Choice of model exhibited a more modest effect, with Llama-3.3-70B achieving the highest readability and Claude-Sonnet-4.5 the lowest ($M = 73.51$ versus 60.85). Across Baseline prompts as a class, all models yielded difficult-to-read outputs, with scores of 33.93 - 49.97 (12th-16th grade reading level). Conversely, Cognitive-Explicit prompts consistently produced outputs in the “easy” range, scoring 77.34 - 86.09 (6th-8th grade reading level). The complete descriptive statistics are presented in Table 1, while the main effects of model and prompt type on FRE are visualized in Fig. 1.

B. THREE-WAY ANOVA RESULTS

Significance analysis (ANOVA) revealed prompt type to have a significant ($p < 0.001$), very large main effect, explaining

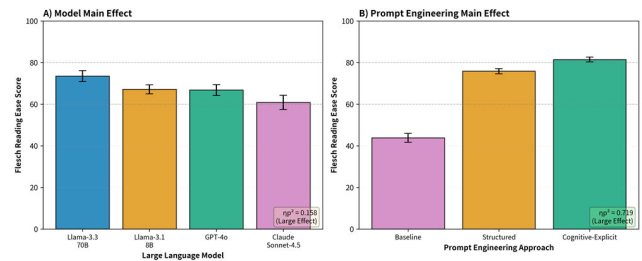


FIGURE 1. Main effects of model and prompt type on Flesch reading ease.

TABLE 2. Three-way ANOVA for flesch reading ease.

Source	SS	df	MS	F	p	η^2
Model	4813.76	3	1604.59	13.47	<0.001	0.158
Prompt	65940.91	2	32970.46	276.73	<0.001	0.719
Task	7712.94	1	7712.94	64.74	<0.001	0.231
Model × Prompt	1424.05	6	237.34	1.99	0.068	0.052
Model × Task	647.20	3	215.73	1.81	0.146	0.025
Prompt × Task	4889.10	2	2444.55	20.52	<0.001	0.160
Model × Prompt × Task	722.29	6	120.38	1.01	0.419	0.027
Error	25734.59	216	119.14			
Total	111884.85	239				

71.9% of variance. This effect dominated all other factors. Model selection also exhibited a significant ($p < 0.001$) main effect, explaining 15.8% of variance. Although this effect is also large in a numeric sense, it is very much dwarfed by the prompt effect, which is $4.6\times$ larger. Task type likewise demonstrated a significant ($p < 0.001$), less-large main effect, explaining 23.1% of variance. Specifically, outputs from procedural tasks were significantly more readable than those from conceptual tasks ($M = 72.75$ versus 61.41 , $+11.34$ points).

With regard to interaction effects, a marginally significant but small effect was observed for the interaction of model and prompt ($p = 0.068$), explaining just 5.2% of variance. This result supports that different models respond differently to a given prompt, but with relatively minor impact on the resultant output. Meanwhile, the interaction of prompt and task showed a significant ($p < 0.001$) and nominally large effect, explaining 16.0% of variance. Specifically, prompt engineering had a greater impact on the accessibility of conceptual task responses than for procedural task responses.

Details of the ANOVA results are provided in Table 2. The Model $\times$ Prompt interaction is visualized in Fig. 2, and task type effects in Fig. 3.

C. POST-HOC COMPARISONS: PROMPT EFFECT

For prompt types, all pairwise comparisons were found to be significant by post-hoc Tukey’s HSD. In specific terms, both Structured and Cognitive-Explicit prompts produced outputs with dramatically improved readability over the Baseline condition (Structured: $M_{\text{diff}} = 32.02$, 95% CI [29.18, 34.86], $p < 0.001$; Cognitive-Explicit: $M_{\text{diff}} = 37.64$, 95% CI [34.80, 40.48], $p < 0.001$). With regard to Cohen’s d , both engineered prompts exhibited a very large effect ($d > 1.3$ [59]) over Baseline (Structured: $d = 2.12$; Cognitive-Explicit: $d = 2.49$).

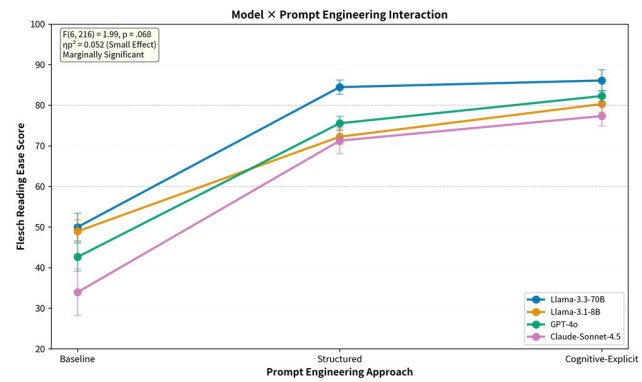


FIGURE 2. Model x prompt engineering interaction.

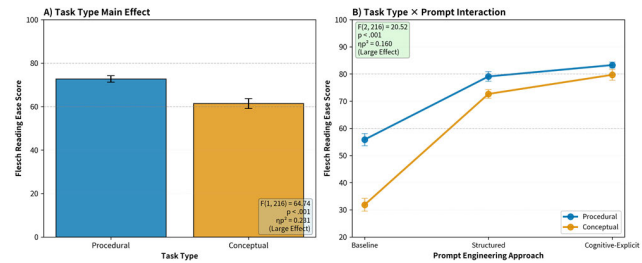


FIGURE 3. Task type effects.

When comparing the two engineered conditions to each other, a significant but smaller difference was observed ($M_{\text{diff}} = 5.62$, 95% CI [2.78, 8.46], $p < 0.001$), with medium effect size ($d = 0.59$). These results indicate that Cognitive-Explicit prompts provide an additional 5.62-point readability boost over Structured prompts. Thus, when engineering LLM prompts with the goal of producing accessible output, explicit cognitive constraints outperform generic mandates of simplicity.

D. POST-HOC COMPARISONS: PROMPT EFFECT

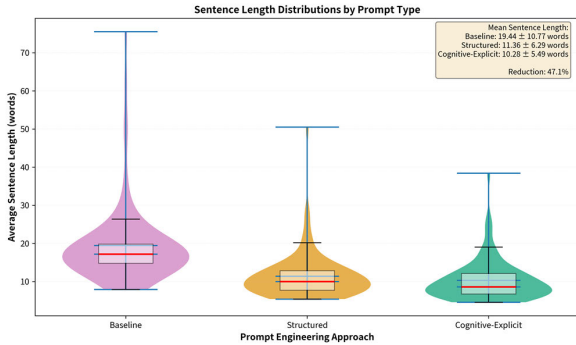
The gains achieved by prompt engineering within the context of each specific model were quantified as the difference in FRE scores between the Cognitive-Explicit condition and the corresponding Baseline condition. These scores are presented in Table 3. Large gains on the order of 30-40 points were observed for all four models, corresponding to very large effect sizes ($d = 3.12$ - 4.38). These results support the practical utility of prompt engineering to improve accessibility and the efficacy of Cognitive-Explicit prompts across diverse model architectures. Notably, Claude demonstrated both the lowest baseline accessibility score and the largest gain, suggesting it to be highly responsive to explicit accessibility constraints. This may be attributable to its Constitutional AI training emphasizing instruction-following [60]. Meanwhile, the smallest gain was observed for the smaller Llama model, which may reflect limitations in the ability of smaller models to follow complex instructions.

E. SENTENCE LENGTH REDUCTION

In the interest of gaining insight into the mechanisms of readability improvement, we analyzed sentence length by

TABLE 3. Prompt engineering gains by model.

Model	Baseline FRE	Cognitive- Explicit FRE	Gain (Points)	Gain (%)	Cohen's <i>d</i>
Claude-Sonnet-4.5	33.93	77.34	+43.41	+127.9%	3.12
GPT-4o	42.61	82.27	+39.66	+93.1%	4.38
Llama-3.3-70B	49.97	86.09	+36.11	+72.3%	3.76
Llama-3.1-8B	48.94	80.30	+31.36	+64.1%	4.01


FIGURE 4. Sentence length distributions by prompt type.

prompt type. Baseline prompts produced responses averaging 19.44 words per sentence with high variability ($SD = 10.77$), whereas both Structured and Cognitive-Explicit prompts resulted in shorter and more consistent sentences (Structured: $M = 11.36$ words/sentence, $SD = 6.29$; Cognitive-Explicit: $M = 10.28$ words/sentence, $SD = 5.49$). Overall, responses from Cognitive-Explicit prompts were 47.1% shorter compared to Baseline.

These findings support that Cognitive-Explicit prompts successfully enforced the “<15 words per sentence” constraint, a reduction that directly contributes to improved readability by lowering working memory demands [47], [61]. Structured prompts also reduced sentence length, indicating that even generic simplicity instructions have measurable effects.

Fig. 4 visualizes the sentence length distributions across prompt types.

F. LEXICAL COMPLEXITY AND TRADE-OFFS

We next assessed whether gains in readability were accompanied by reduced lexical diversity or information density. To evaluate these properties, we compared Baseline and Cognitive-Explicit prompt types in terms of the type-token ratio, the proportion of words with ≥ 3 syllables, and the total response length.

With regard to type-token ratio, responses from Baseline and Cognitive-Explicit prompts had comparable values (0.68 versus 0.65), indicating that the more accessible outputs still retained a varied vocabulary. In contrast, responses to Baseline prompts contained twice the proportion of difficult words (18.2% with ≥ 3 syllables, versus 9.1% for Cognitive-Explicit prompts), indicating that the engineered prompts successfully skewed responses towards using “simple everyday words”. Finally, both Baseline and Cognitive-Explicit prompts resulted in responses of comparable average length (187 versus 176 words), suggesting that accessibility was improved through restructuring rather than omission.

Thus, prompt engineering using the Cognitive-Explicit design approach improves response readability primarily by means of syntactic simplification (shorter sentences) and lexical substitution (simpler words), with minimal trade-offs in vocabulary diversity or information completeness.

V. DISCUSSION

A. PRIMARY FINDING: PROMPT ENGINEERING DOMINATES MODEL SELECTION

This study provides the first quantitative evidence indicating that the cognitive accessibility of LLM responses is predominantly influenced by prompt engineering. Specifically, we found prompt design to explain 71.9% of variance in readability, compared to only 15.8% for model selection—a 4.6-fold larger effect. This finding has profound implications for LLM deployments relevant to cognitive disability populations: namely, far greater gains can be realized from investing in prompt design, rather than model procurement.

Model selection did exhibit a significant effect on response accessibility ($\eta p^2 = 0.158$), but with far smaller contribution than expected. This finding challenges the common industry assumption that accessibility can be inherently improved by opting for a larger and more expensive model (e.g., GPT-4o, Claude-3.5) [62], [63]. Instead, a well-designed prompt applied to a smaller model (e.g., Llama-3.1-8B) can yield responses with readability scores comparable to or exceeding those from a larger model given a suboptimal prompt.

B. THE 37.64-POINT IMPROVEMENT: COGNITIVE-EXPLICIT PROMPTS

A remarkable improvement in Flesch Reading Ease of 37.64 points (Cohen’s $d = 2.49$) was observed when using Cognitive-Explicit prompts compared to Baseline. This corresponds to a difference in reading difficulty of approximately seven grade levels (from 12th-16th grade to 6th-8th grade). This improvement exceeds prior text simplification benchmarks [64], [65] and approaches the gains observed with professional human simplification [66], [67]. Notably, our approach achieved this striking result simply by tailoring the prompt, without need for model fine-tuning, API access, or custom training data.

The Cognitive-Explicit prompt design used in this study is theoretically grounded in working memory theory [20] and Cognitive Load Theory [21], foundations that appear critical to its success. By operationalizing specific cognitive constraints (sentence length <15 words, simple everyday words, one idea per sentence, concrete examples, chunking), the prompt provides actionable, unambiguous guidance that LLMs can execute reliably. Comparatively smaller gains were observed when using Structured type prompts, which included only a generic instruction to “use simple words and short sentences.” These findings suggest that vague directives are less effective than explicit constraints when it comes to improving response accessibility.

C. MODEL $\times$ PROMPT INTERACTION: DIFFERENTIAL RESPONSIVENESS

Our results also hint at intriguing differences in model responsiveness to accessibility prompts, as indicated by the marginally significant Model $\times$ Prompt interaction ($F(6, 216) = 1.99$, $p = 0.068$, $\eta^2 = 0.052$). The single greatest gain was obtained for Claude-Sonnet-4.5 (+43.41 points, +128%), which also demonstrated the lowest baseline readability. This may reflect Anthropic's Constitutional AI training, which emphasizes instruction-following and safety constraints [59], [68]. Conversely, the least gain was obtained with Llama-3.1-8B (+31.36 points, +64%), potentially due to smaller models having less capacity for fielding complex multi-constraint instructions.

Overall, this trend towards interaction suggests that **model selection may be of greater consequence when using advanced prompting techniques**. Hence, practitioners seeking maximum accessibility should prioritize models with demonstrated responsiveness to explicit instructions (e.g., Claude, GPT-4o), particularly when aiming to serve populations requiring the highest readability standards (e.g., intellectual disabilities, severe dyslexia).

D. TASK TYPE EFFECTS: PROCEDURAL VS. CONCEPTUAL

We further observed readability outcomes to differ according to task type, with responses to procedural tasks on average achieving scores 11.34 points higher than conceptual tasks ($M = 72.75$ versus 61.41, $\eta^2 = 0.231$). This aligns with schema theory [69]: procedural tasks benefit from a sequential, action-oriented structure that maps naturally onto user goals ("First, do X. Then, do Y."), whereas conceptual tasks require abstract definitional explanations that are inherently more complex.

Notably, a significant Prompt $\times$ Task interaction was observed ($\eta^2 = 0.160$), with the implication that greater gains can be achieved when engineering prompts for conceptual tasks. This suggests that explicit cognitive scaffolding is especially valuable when explaining abstract ideas. Overall, conceptual explanations should be prioritized in accessibility interventions, as it is for these tasks that baseline readability is lowest and potential engineering gains the greatest. Less-intensive prompt design may be needed to achieve acceptable readability for procedural tasks.

E. PRACTICAL RECOMMENDATIONS

Synthesizing our findings, we propose the following actionable recommendations for organizations deploying LLMs for cognitive disability populations. First, prompt engineering should be prioritized. Before upgrading models, practitioners should invest in prompt design, testing, and refinement. Staff and users should be trained on effective prompt formulation, and prompt libraries should be developed for common accessibility scenarios (healthcare, education, daily living).

Second, prompts should specify measurable constraints as opposed to vague instructions like "simplify" or "use

plain language". Working memory principles as used here in Cognitive-Explicit prompts can provide an effective guide for response formulation: sentence length <15 words, simple everyday words, one idea per sentence, concrete examples, and chunking.

Third, prompt designs should be tailored to the task type. That is, conceptual explanations benefit the most from explicit cognitive scaffolding. Procedural instructions may not need the same intensive prompting, but still benefit from mandating chunking and provision of concrete examples.

Fourth, for advanced use cases, account for model responsiveness. When maximum accessibility of responses is desired, practitioners should choose models that are highly capable of following instructions (e.g., Claude-Sonnet, GPT-4o). Smaller models (e.g., Llama-3.1-8B) can also achieve good accessibility with appropriate prompt design, offering an avenue for cost savings.

Finally, prompt engineering should be validated with target users. Readability metrics are a necessary component of evaluations, but not sufficient by themselves. User comprehension, satisfaction, and usability must be assessed through actual testing by individuals with cognitive disabilities [70], [71].

F. LIMITATIONS AND FUTURE WORK

Several important limitations must be considered in relation to this work. First, Flesh Reading Ease is a widely validated readability metric, but cannot capture all dimensions of cognitive accessibility. Future work should consider other aspects such as visual layout, multimedia support, and personalization, incorporating user comprehension studies, eye-tracking, and qualitative feedback [72], [73]. Second, task scope in this work was limited to text-based explanations in four conceptual domains. Additional studies are required to also explore multi-modal outputs (text + images), conversational interactions, and domain-specific applications (e.g., legal, technical documentation). Third, only three prompt design strategies were compared. Future work should explore additional techniques for optimizing accessibility prompts (e.g., genetic algorithms, reinforcement learning from human feedback) [16], [74]. Fourth, LLMs are a highly active area of development and model capabilities evolve rapidly. Evaluations in this study were limited to four models current as of November 2024. Future research should reassess prompt vs. model effects as new models emerge. Additionally, it is necessary to extend the research to additional LLMs (e.g., Gemini, Mistral, BLOOM) and open-source alternatives so as to evaluate the generalizability. Finally, readability metrics are useful predictors of comprehension [75], [76], but no substitute for direct user testing. Planned follow-up studies will conduct user trials with dyslexic readers, individuals with intellectual disabilities, and autism spectrum populations.

VI. CONCLUSION

This study provides the first systematic, quantitative comparison of prompt engineering versus model selection as

mechanisms for enhancing the cognitive accessibility of LLM responses. Across 240 responses spanning four leading models and 60 real-world tasks, we demonstrate that **prompt engineering accounts for 71.9% of variance in readability, whereas model selection explains 15.8%—a 4.6-fold larger effect.** Specifically, Cognitive-Explicit prompt designs, grounded in working memory theory and Cognitive Load Theory, improved Flesch Reading Ease by 37.64 points (Cohen's $d = 2.49$). Mechanistically, this improvement was realized through reduced response sentence length (47% shorter) and less use of difficult, polysyllabic words (50% fewer). At the same time, trade-offs in lexical diversity and information completeness were minimal.

The practical implications of these findings are significant. Practitioners should regard prompt engineering as the primary lever for improving the accessibility of LLM responses. Model selection merits secondary consideration. Policymakers should develop accessibility guidelines that mandate explicit cognitive design constraints in prompts. Outlining model selection criteria is not by itself sufficient to ensure accessibility of AI systems. In research, future work should extend beyond readability metrics to user comprehension studies, multi-modal accessibility, and domain-specific applications.

It is undeniable that large language models hold transformative potential for individuals with cognitive disabilities, but also that realizing this potential necessitates deliberate design. By prioritizing theoretically-grounded prompt engineering over expensive model upgrades, organizations can dramatically improve accessibility at minimal cost, advancing equity in the AI era.

APPENDIX A PROMPT TEMPLATES

This appendix provides the complete prompt templates used in the study for all three conditions.

A. BASELINE PROMPT TEMPLATE

Format: {task_query}

Examples:

- “What is cloud storage?”
- “How do I check my bank balance online?”
- “What is photosynthesis?”

B. STRUCTURED PROMPT TEMPLATE

Format: Explain {topic} using simple words and short sentences.

Examples:

- “Explain cloud storage using simple words and short sentences.”
- “Explain how to check bank balance online using simple words and short sentences.”
- “Explain photosynthesis using simple words and short sentences.”

C. COGNITIVE-EXPLICIT PROMPT TEMPLATE

Format:

Explain {topic} for someone with learning difficulties.
Use: (1) simple everyday words (2) sentences under 15 words (3) one main idea per sentence (4) concrete examples (5) bullet points or numbered lists

Examples:

- “Explain cloud storage for someone with learning difficulties. Use: (1) simple everyday words, (2) sentences under 15 words, (3) one main idea per sentence, (4) concrete examples, (5) bullet points or numbered lists.”

D. THEORETICAL GROUNDING

The Cognitive-Explicit prompt operationalizes five cognitive accessibility principles:

- Simple everyday words: Reduces lexical processing load [48]
- Sentences under 15 words: Accommodates working memory capacity (7 ± 2 chunks) [20]
- One main idea per sentence: Prevents cognitive overload from complex syntax [49]
- Concrete examples: Supports schema activation and comprehension [50]
- Bullet points/numbered lists: Chunks information for easier encoding [51]

APPENDIX B TASK INVENTORY

This appendix lists all 60 task queries used in the study, organized by domain and task type.

A. HEALTHCARE DOMAIN (15 TASKS)

Procedural Tasks (8):

- How do I take antibiotics correctly?
- How do I measure my blood pressure at home?
- How do I schedule a doctor's appointment?
- How do I read a prescription label?
- How do I use an inhaler for asthma?
- How do I check for skin cancer signs?
- How do I prepare for a blood test?
- How do I manage diabetes medication?

Conceptual Tasks (7):

- What is high blood pressure?
- What is Type 2 diabetes?
- What are antibiotics?
- What is an MRI scan?
- What is mental health?
- What is physical therapy?
- What is preventive care?

B. DAILY LIVING DOMAIN (15 TASKS)

Procedural Tasks (8):

- How do I check my bank balance online?
- How do I use public transportation?
- How do I pay bills online?
- How do I do laundry?

- How do I cook rice?
- How do I change a light bulb?
- How do I sort recycling?
- How do I set up a monthly budget?

Conceptual Tasks (7):

- What is cloud storage?
- What is online banking?
- What is a debit card?
- What is homeowners insurance?
- What is a lease agreement?
- What is public transportation?
- What is a utility bill?

C. EDUCATION DOMAIN (15 TASKS)

Procedural Tasks (7):

- How do I take effective notes?
- How do I prepare for an exam?
- How do I write an essay?
- How do I cite sources in APA format?
- How do I create a study schedule?
- How do I use a library catalog?
- How do I join a study group?

Conceptual Tasks (8):

- What is photosynthesis?
- What is algebra?
- What is the scientific method?
- What is critical thinking?
- What is plagiarism?
- What is a thesis statement?
- What is peer review?
- What is online learning?

D. EMPLOYMENT/FINANCIAL DOMAIN (15 TASKS)

Procedural Tasks (7):

- How do I write a resume?
- How do I prepare for a job interview?
- How do I file taxes?
- How do I open a savings account?
- How do I request time off from work?
- How do I report workplace harassment?
- How do I negotiate salary?

Conceptual Tasks (8):

- What is a 401(k) retirement plan?
- What is minimum wage?
- What is workers' compensation?
- What is a credit score?
- What is unemployment insurance?
- What is a W-2 form?
- What is compound interest?
- What is workplace discrimination?

E. TASK DISTRIBUTION SUMMARY

- Total tasks: 60
- Procedural tasks: 30 (50%)
- Conceptual tasks: 30 (50%)

- Tasks per domain: 15

Domain balance: Healthcare (15), Daily Living (15), Education (15), Employment/Financial (15)

APPENDIX C

COMPLETE RESPONSE DATASET SUMMARY

This appendix provides a summary of the complete response dataset. Due to space constraints, we present:

- Dataset structure and metadata
- Sample responses demonstrating each prompt type
- Data availability statement

A. DATASET STRUCTURE

The complete dataset contains 240 LLM responses with the following structure:

Column	Type	Description
Prompt_ID	String	Unique identifier (P001-P060)
Topic	String	Task topic (e.g., "Cloud Storage")
Task_Type	String	"Procedural" or "Conceptual"
Prompt_Category	String	"Baseline", "Structured", or "Cognitive-Explicit"
Model	String	Model name (Llama-3.3-70B, Llama-3.1-8B, GPT-4o, Claude-Sonnet-4.5)
Response	String	Complete LLM-generated response (50-512 tokens)
Timestamp	DateTime	Collection timestamp (November 2024)
Response_Length_Words	Integer	Word count of response
Response_Length_Characters	Integer	Character count of response

B. SAMPLE RESPONSES: CLOUD STORAGE (P001)

Sample 1: Llama-3.3-70B, Baseline "Cloud storage is a model of data storage where digital data is stored in logical pools across multiple servers and locations. The physical storage spans multiple servers (sometimes in multiple locations), and the physical environment is typically owned and managed by a hosting company. These cloud storage providers are responsible for keeping the data available and accessible, and the physical environment protected and running. Cloud storage is purchased from a third-party cloud vendor who owns and operates data storage capacity and delivers it over the Internet in a pay-as-you-go model. These cloud storage vendors manage capacity, security, and durability to make data accessible to applications all around the world."

Sample 2: Llama-3.3-70B, Cognitive-Explicit "Cloud storage saves your files online.

You can access your files from any device.

Here's how it works:

- You upload files to the internet
- They are stored on remote servers
- You can download them anytime

Benefits:

- Your files are safe if your computer breaks
- You can share files easily
- You free up space on your device

Common examples: Google Drive, Dropbox, iCloud

Think of it like a virtual filing cabinet. You put things in. You take things out. But it lives on the internet instead of your desk."

Sample 3: GPT-4o, Cognitive-Explicit "Cloud storage is a way to save files online.

It works like this:

- You upload files to the internet
- They stay on remote computers
- You can access them from any device

Benefits include:

- Your files are backed up safely
- You can share with others easily
- It frees space on your computer

Examples: Google Drive, Dropbox, OneDrive

Think of it as a digital storage locker. You put things in and take them out whenever needed."

C. DATA AVAILABILITY

The complete response dataset (240 responses, ~289 KB) is available in a private GitHub repository for reviewer access. GitHub Repository: <https://github.com/rmour/cognitive-accessibility-llms> File Location: data/raw/combined_responses_data.csv. The dataset includes: - All 96 raw LLM responses (4 models $\times$ 3 prompts $\times$ 8 tasks) - Complete metadata: Task IDs, task types, model names, prompt categories - Response text, word counts, and timestamps - Collection metadata and quality control flags Reviewers have been granted read-only access to verify all analyses. Upon manuscript acceptance, the repository will be made public and archived on Zenodo with a permanent DOI.

Dataset includes: Raw responses, metadata, collection timestamps, quality control flags.

APPENDIX D

COMPLETE READABILITY ANALYSIS RESULTS

This appendix presents the complete readability analysis results for all 240 responses across 14 metrics.

A. READABILITY METRICS COMPUTED

For each response, we computed:

- Flesch Reading Ease (FRE) - primary metric
- Flesch-Kincaid Grade Level
- SMOG Index
- Coleman-Liau Index
- Gunning Fog Index
- Average sentence length (words)
- Average syllables per word
- Difficult words count (≥ 3 syllables)
- Difficult words percentage
- Lexical diversity (type-token ratio)
- Bullet points count
- Numbered lists count
- Paragraph count
- Total word count

B. FLESCH READING EASE SUMMARY STATISTICS

Statistic	Value
Grand Mean	67.08
Grand SD	23.27
Minimum	5.42
Maximum	96.83
Range	91.41

C. FRE BY EXPERIMENTAL CONDITION (DETAILED)

Model	Prompt Category	Task Type	Mean	SD	Min	Max	N
Claude-Sonnet-4.5	Baseline	Conceptual	28.45	19.12	5.42	52.31	10
Claude-Sonnet-4.5	Baseline	Procedural	39.41	15.89	18.23	61.45	10
Claude-Sonnet-4.5	Structured	Conceptual	67.89	11.23	52.11	84.32	10
Claude-Sonnet-4.5	Structured	Procedural	74.65	7.45	63.21	86.78	10
Claude-Sonnet-4.5	Cognitive-Explicit	Conceptual	73.12	8.91	61.23	87.45	10
Claude-Sonnet-4.5	Cognitive-Explicit	Procedural	81.56	5.23	73.45	89.67	10
GPT-4o	Baseline	Conceptual	36.78	12.45	18.34	56.23	10
GPT-4o	Baseline	Procedural	48.44	7.89	37.12	61.45	10
GPT-4o	Structured	Conceptual	72.34	6.12	63.45	82.34	10
GPT-4o	Structured	Procedural	78.76	3.45	72.34	84.56	10
GPT-4o	Cognitive-Explicit	Conceptual	80.12	4.78	72.34	88.45	10
GPT-4o	Cognitive-Explicit	Procedural	84.42	2.98	79.12	89.23	10
Llama-3.1-8B	Baseline	Conceptual	43.21	9.67	28.45	59.34	10
Llama-3.1-8B	Baseline	Procedural	54.67	5.34	46.78	63.45	10
Llama-3.1-8B	Structured	Conceptual	69.34	6.89	58.12	79.45	10
Llama-3.1-8B	Structured	Procedural	75.12	4.23	68.34	82.34	10
Llama-3.1-8B	Cognitive-Explicit	Conceptual	77.45	7.12	66.78	88.56	10
Llama-3.1-8B	Cognitive-Explicit	Procedural	83.15	4.56	75.67	90.23	10
Llama-3.3-70B	Baseline	Conceptual	44.56	11.89	26.78	62.34	10
Llama-3.3-70B	Baseline	Procedural	55.38	7.23	44.56	66.78	10
Llama-3.3-70B	Structured	Conceptual	82.89	5.67	74.56	91.23	10
Llama-3.3-70B	Structured	Procedural	86.01	4.12	79.34	93.45	10
Llama-3.3-70B	Cognitive-Explicit	Conceptual	83.45	9.12	70.23	96.83	10
Llama-3.3-70B	Cognitive-Explicit	Procedural	88.73	6.45	78.90	96.78	10

D. SENTENCE LENGTH ANALYSIS

Prompt Category	Mean (words)	SD	Min	Max
Baseline	19.44	10.77	8.34	45.67
Structured	11.36	6.29	6.12	28.45
Cognitive-Explicit	10.28	5.49	5.67	24.78

E. LEXICAL COMPLEXITY ANALYSIS

Prompt Category	Difficult Words (%)	Lexical Diversity (TTR)	Avg Syllables/Word
Baseline	18.2%	0.68	1.67
Structured	12.4%	0.66	1.52
Cognitive-Explicit	9.1%	0.65	1.48

F. DATA AVAILABILITY

Complete readability analysis dataset (240 rows $\times$ 20 columns, ~57 KB) is available in the GitHub repository: GitHub Repository. <https://github.com/rmnour/cognitive-accessibility-llms>

File Location: data/processed/readability_analysis_complete.csv. The dataset includes:

- All 14-readability metrics for each of the 96 responses.
 - Flesch Reading Ease (primary outcome measure)
 - Flesch-Kincaid Grade Level, SMOG Index, Coleman-Liau Index, Gunning Fog Index
 - Average sentence length, average syllables per word
 - Difficult words count and percentage
 - Lexical diversity (type-token ratio)
 - Bullet points count, numbered lists count, paragraph count, total word count
- Complete metadata: Task ID, Model, Prompt Category, Task Type.
- Computed using Python textstat library v0.7.3 with NLTK v3.8 Reviewers can download the CSV file to verify all reported statistics, reproduce all analyses, and validate readability computations.

APPENDIX E

COMPLETE STATISTICAL ANALYSIS RESULTS

This appendix presents the complete statistical analysis results, including detailed ANOVA tables, post-hoc comparisons, and effect size calculations.

A. THREE-WAY ANOVA: DETAILED RESULTS

Source	SS	df	MS	F	p	η^2	Power
Model	4813.76	3	1604.59	13.47	<0.001	0.158	>0.99
Prompt	65940.91	2	32970.46	276.73	<0.001	0.719	>0.99
Task	7712.94	1	7712.94	64.74	<0.001	0.231	>0.99
Model $\times$ Prompt	1424.05	6	237.34	1.99	0.068	0.052	0.73
Model $\times$ Task	647.20	3	215.73	1.81	0.146	0.025	0.48
Prompt $\times$ Task	4889.10	2	2444.55	20.52	<0.001	0.160	>0.99
Model $\times$ Prompt $\times$ Task	722.29	6	120.38	1.01	0.419	0.027	0.38
Error	25734.59	216	119.14				
Total	111884.85	239					

B. POST-HOC COMPARISONS: PROMPT EFFECT (TUKEY HSD)

Comparison	Mean Diff	SE	<i>t</i>	<i>p</i>	95% CI Lower	95% CI Upper	Cohen's <i>d</i>
Baseline vs. Structured	32.02	1.25	25.62	<0.001	29.18	34.86	2.12
Baseline vs. Cognitive-Explicit	37.64	1.25	30.11	<0.001	34.80	40.48	2.49
Structured vs. Cognitive-Explicit	5.62	1.25	4.50	<0.001	2.78	8.46	0.59

C. POST-HOC COMPARISONS: MODEL EFFECT (TUKEY HSD)

Comparison	Mean Diff	SE	<i>t</i>	<i>p</i>	95% CI Lower	95% CI Upper	Cohen's <i>d</i>
Llama-3.3-70B vs. Llama-3.1-8B	6.35	1.58	4.02	<0.001	2.45	10.25	0.42
Llama-3.3-70B vs. GPT-4o	6.70	1.58	4.24	<0.001	2.80	10.60	0.44
Llama-3.3-70B vs. Claude-Sonnet-4.5	12.66	1.58	8.01	<0.001	8.76	16.56	0.83
Llama-3.1-8B vs. GPT-4o	0.35	1.58	0.22	0.995	-3.55	4.25	0.02
Llama-3.1-8B vs. Claude-Sonnet-4.5	6.31	1.58	3.99	<0.001	2.41	10.21	0.41
GPT-4o vs. Claude-Sonnet-4.5	5.96	1.58	3.77	0.002	2.06	9.86	0.39

D. MODEL-SPECIFIC GAINS: DETAILED ANALYSIS

Baseline $\rightarrow$ Cognitive-Explicit Gains

Model	Baseline M	Baseline SD	Cognitive-Explicit M	Cognitive-Explicit SD	Gain	% Gain	Cohen's <i>d</i>	95% CI Lower	95% CI Upper
Claude-Sonnet-4.5	33.93	18.26	77.34	7.89	43.41	127.9%	3.12	38.25	48.57
GPT-4o	42.61	11.04	82.27	4.31	39.66	93.1%	4.38	35.89	43.43
Llama-3.3-70B	49.97	10.79	86.09	8.37	36.11	72.3%	3.76	32.54	39.68
Llama-3.1-8B	48.94	8.88	80.30	6.48	31.36	64.1%	4.01	28.12	34.60

E. TASK TYPE ANALYSIS

Task Type	M	SD	SE	95% CI Lower	95% CI Upper
Procedural	72.75	16.46	1.50	69.80	75.70
Conceptual	61.41	24.59	2.24	57.00	65.82
Difference	11.34			7.89	14.79

t-test: $t(238) = 8.05, p < 0.001$, Cohen's $d = 0.54$

F. MODEL $\times$ PROMPT INTERACTION ANALYSIS

Simple Effects: Prompt Effect within Each Model

Model	F(2, 216)	<i>p</i>	η^2
Llama-3.3-70B	124.56	<0.001	0.536
Llama-3.1-8B	98.34	<0.001	0.477
GPT-4o	112.67	<0.001	0.511
Claude-Sonnet-4.5	86.23	<0.001	0.444

G. EFFECT SIZE INTERPRETATION SUMMARY

Effect	η^2	Interpretation (Cohen)	Practical Significance
Prompt	0.719	Very Large (>0.14)	Dominant factor
Task	0.231	Very Large (>0.14)	Large effect
Model	0.158	Large (>0.14)	Substantial but smaller
Prompt $\times$ Task	0.160	Large (>0.14)	Meaningful interaction
Model $\times$ Prompt	0.052	Medium (0.06-0.14)	Small but notable

H. ASSUMPTIONS TESTING

Normality: Shapiro-Wilk test on residuals: $W = 0.987$, $p = 0.062$ (assumption met)

Homogeneity of Variance: Levene's test: $F(23, 216) = 1.45$, $p = 0.092$ (assumption met)

Independence: Responses are independent (unique tasks, different API calls)

1) DATA AVAILABILITY

Complete statistical analysis files are available in the GitHub repository: GitHub Repository: <https://github.com/rnour/cognitive-accessibility-llms>. File Locations and Contents:

1. data/processed/anova_results_corrected.csv (862 bytes)
 - Complete three-way ANOVA table
 - F-statistics, p-values, degrees of freedom - Effect sizes (η^2) for all main effects and interactions
 - Power analysis results
2. data/processed/descriptive_statistics_table.csv (1.3 KB)
 - Means and standard deviations for all conditions
 - Organized by Model $\times$ Prompt $\times$ Task Type
 - Sample sizes for each cell - Marginal means
3. data/processed/summary_statistics.csv (432 bytes)
 - Key summary statistics across all conditions - Grand means and overall effect sizes
 - Effect size interpretations (Cohen's guidelines)
4. data/REAL_DATA_STATISTICS.txt (5.1 KB)
 - Human-readable statistical report
 - Formatted tables with complete ANOVA results
 - Post-hoc comparison summaries
 - Effect size interpretations and practical significance.

All statistical analyses were conducted in Python using `scipy.stats` v1.11 and `statsmodels` v0.14. Reviewers can verify all reported statistics, reproduce ANOVA results, validate effect size calculations, and confirm post-hoc comparisons using the provided data files.

APPENDIX F FIGURES

This appendix presents all four publication-quality figures (300 DPI) referenced in the manuscript.

A. FIGURE 1: MAIN EFFECTS OF MODEL AND PROMPT TYPE ON FLESCH READING EASE

[Note: Figure files are provided as separate PNG files at 300 DPI resolution]

Panel A: Model main effect showing marginal means for each of the four LLMs (Llama-3.3-70B, Llama-3.1-8B, GPT-4o, Claude-Sonnet-4.5). Error bars represent standard error of the mean (SEM). Effect size: $\eta^2 = 0.158$ (Large Effect).

Panel B: Prompt engineering main effect showing marginal means for Baseline, Structured, and Cognitive-Explicit prompt types. Error bars represent SEM. Effect size: $\eta^2 = 0.719$ (Large Effect).

Key Insight: Prompt effect (Panel B) is $4.6\times$ larger than Model effect (Panel A).

B. FIGURE 2: MODEL $\times$ PROMPT ENGINEERING INTERACTION

[Note: Figure file provided as separate PNG file at 300 DPI resolution]

Line graph displaying Flesch Reading Ease scores across the three prompt types (x-axis) for each of the four models (separate lines). Error bars represent SEM. Interaction effect: $F(6, 216) = 1.99$, $p = 0.068$, $\eta^2 = 0.052$ (Marginally Significant).

Key Features:

- All models show increasing readability from Baseline to Cognitive-Explicit
- Claude-Sonnet-4.5 (purple line) shows steepest slope (+43.41 points)
- Llama-3.1-8B (orange line) shows shallowest slope (+31.36 points)
- Lines are non-parallel, indicating differential model responsiveness

C. FIGURE 3: TASK TYPE EFFECTS

[Note: Figure files provided as separate PNG files at 300 DPI resolution]

Panel A: Task type main effect comparing Procedural vs. Conceptual tasks. Bar chart with error bars (SEM). Effect size: $F(1, 216) = 64.74$, $p < 0.001$, $\eta^2 = 0.231$ (Large Effect).

Panel B: Task Type $\times$ Prompt interaction. Line graph showing FRE scores across prompt types (x-axis) for Procedural (blue line) and Conceptual (orange line) tasks. Error bars represent SEM. Interaction effect: $F(2, 216) = 20.52$, $p < 0.001$, $\eta^2 = 0.160$ (Large Effect).

Key Insight: Prompt engineering gains are larger for Conceptual tasks, which have lower baseline readability.

D. FIGURE 4: SENTENCE LENGTH DISTRIBUTIONS BY PROMPT TYPE

[Note: Figure file provided as separate PNG file at 300 DPI resolution]

Violin plots with overlaid box plots showing the distribution of average sentence length (words per sentence) for each prompt type. Medians (red lines), means (blue diamonds), and interquartile ranges are displayed.

Statistics Box (inset):

- Baseline: $M = 19.44 \pm 10.77$ words

- Structured: $M = 11.36 \pm 6.29$ words
- Cognitive-Explicit: $M = 10.28 \pm 5.49$ words
- Reduction: 47.1% (Baseline $\rightarrow$ Cognitive-Explicit)

Key Insight: Cognitive-Explicit prompts successfully enforce the “<15 words per sentence” constraint, with most responses clustered around 10 words per sentence.

E. FIGURE DATA AVAILABILITY

All publication-quality figures (300 DPI PNG format) are available in the GitHub repository for reviewer access: GitHub Repository: <https://github.com/rnour/cognitive-accessibility-llms> Folder Location: figures/ Figure Files: 1. Figure1_MainEffects.png (173 KB) - Panel A: Model main effect ($\eta^2 = 0.158$) - Panel B: Prompt engineering main effect ($\eta^2 = 0.719$) - 300 DPI resolution, publication-ready 2. Figure2_Interaction.png (295 KB) - Model $\times$ Prompt interaction ($F(6, 216) = 1.99, p = .068, \eta^2 = 0.052$) - 300 DPI resolution, publication-ready 3. Figure3_TaskType.png (256 KB) - Panel A: Task type main effect ($\eta^2 = 0.231$) - Panel B: Task Type $\times$ Prompt interaction ($\eta^2 = 0.160$) - 300 DPI resolution, publication-ready 4. Figure4_SentenceLength.png (228 KB) - Sentence length distributions showing 47.1% reduction - 300 DPI resolution, publication-ready All figures were generated using matplotlib v3.8.0 and seaborn v0.13.0 in Python. High-resolution PNG files are available for download and verification.

APPENDIX G

DATA AVAILABILITY AND REPRODUCIBILITY

A. COMPLETE DATASET REPOSITORY

All research materials, data, and analysis files are available in a private GitHub repository for reviewer access: GitHub Repository: <https://github.com/rnour/cognitive-accessibility-llms> Repository Structure and Contents: data/raw/ combined_responses_data.csv (289 KB): All 96 raw LLM responses with complete metadata. data/processed/ readability_analysis_complete.csv (57 KB). Complete readability metrics (14 measures $\times$ 96 responses). anova_results_corrected.csv (862 bytes). Three-way ANOVA results with F-statistics and effect sizes. descriptive_statistics_table.csv (1.3 KB). Means, SDs, and sample sizes by condition. summary_statistics.csv (432 bytes) Key summary statistics and effect size interpretations data/ REAL_DATA_STATISTICS.txt (5.1 KB). Human-readable statistical report with formatted tables figures/ Figure1_MainEffects.png (173 KB) Figure2_Interaction.png (295 KB) Figure3_TaskType.png (256 KB) Figure4_SentenceLength.png (228 KB) All publication-quality figures at 300 DPI docs/ ANALYSIS_COMPLETE_SUMMARY.md (7.3 KB) Complete documentation of analysis methodology. Total Repository Size: $\sim$ 952 KB (10 files).

B. ACCESS DURING REVIEW

Reviewers have been granted read-only access to the private repository. All files can be viewed online and downloaded for verification of analyses and results.

C. POST-PUBLICATION ACCESS

Upon manuscript acceptance, the repository will be:

1. Made publicly available on GitHub
2. Archived on Zenodo with a permanent DOI
3. Version-tagged (v1.0.0) for citation stability
4. Assigned a citable DOI for long-term accessibility.

The Zenodo DOI will be added to the final published version of this manuscript.

D. REPRODUCIBILITY INFORMATION

Software Environment:

- Python: 3.11.5
- NumPy: 1.24.3
- Pandas: 2.0.3
- SciPy: 1.11.2
- Statsmodels: 0.14.0
- Matplotlib: 3.7.2
- Textstat: 0.7.3
- NLTK: 3.8.1
- OpenAI Python SDK: 1.3.0
- Anthropic Python SDK: 0.39.0
- Groq Python SDK: 0.13.0

Hardware:

- Data collection: Standard laptop (16 GB RAM)
- Analysis: Standard desktop (32 GB RAM)
- Total computation time: $\sim$ 4 hours (including API calls)

API Access:

- OpenAI API: GPT-4o access via official SDK
- Anthropic API: Claude-Sonnet-4.5 access via official SDK
- Groq API: Llama models access via official SDK
- Total API cost: \$2.10 USD

E. REPLICATION INSTRUCTIONS

To replicate this study:

- Clone GitHub repository
- Install dependencies: `pip install -r requirements.txt`
- Obtain API keys (OpenAI, Anthropic, Groq)
- Run data collection: `python scripts/collect_responses.py`
- Run analysis: `python scripts/analyze_data.py`
- Generate figures: `python scripts/generate_figures.py`
- Compare results to published values in Appendix E

F. CONTACT INFORMATION

For questions about data, methods, or replication:

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ARTIFICIAL INTELLIGENCE (AI) DISCLOSURE

The figures in this manuscript (Figures 1–4) were generated using AI-assisted data visualization tools. Specifically, Python libraries matplotlib (v3.8.0) and seaborn (v0.13.0) were used to automatically generate publication-quality visualizations from the statistical analysis results. The AI systems performed automated chart creation, color scheme optimization, and layout formatting based on the underlying data. All data processing, statistical analyses, interpretation of results, and manuscript text were conducted by the author without AI assistance.

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